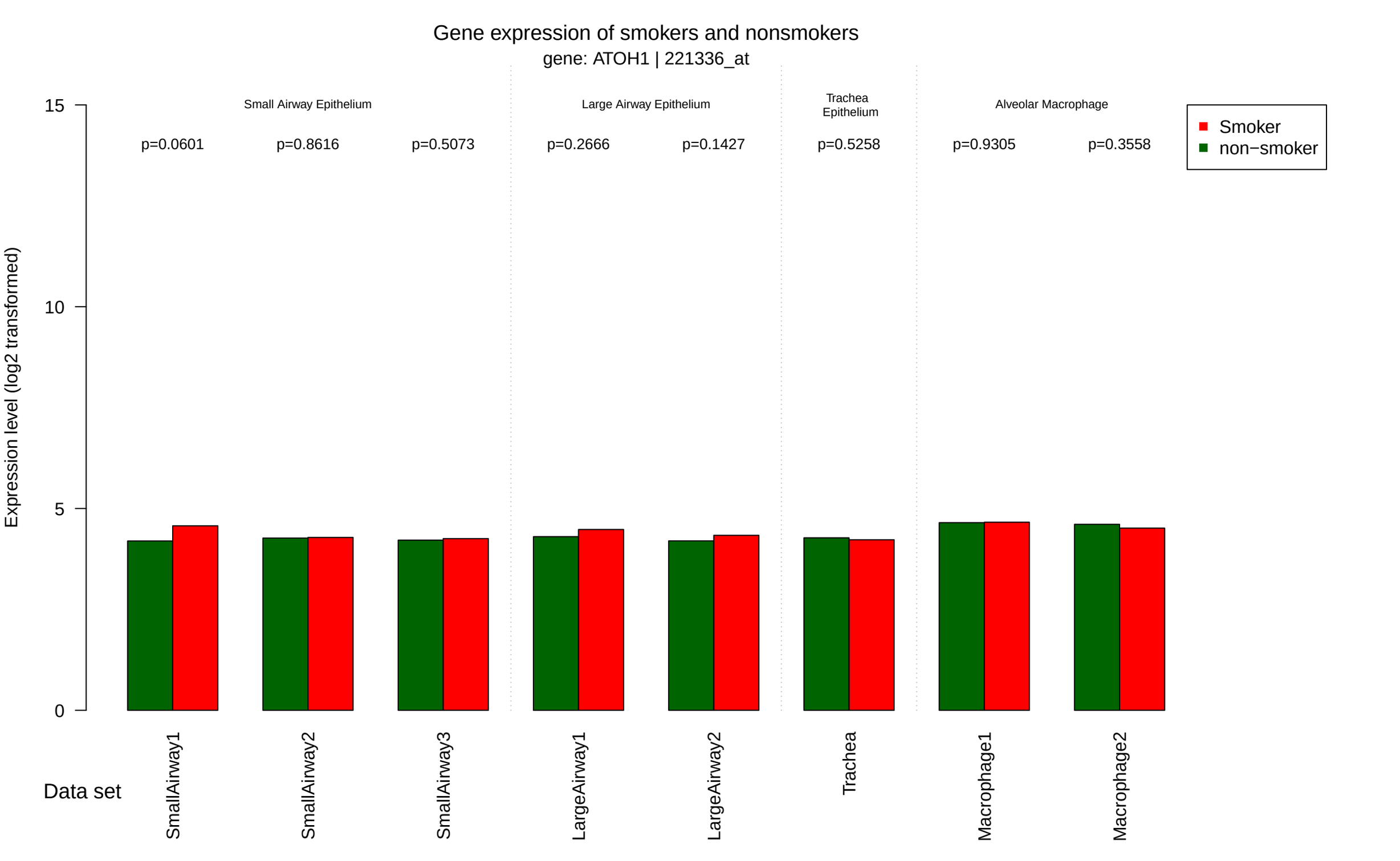
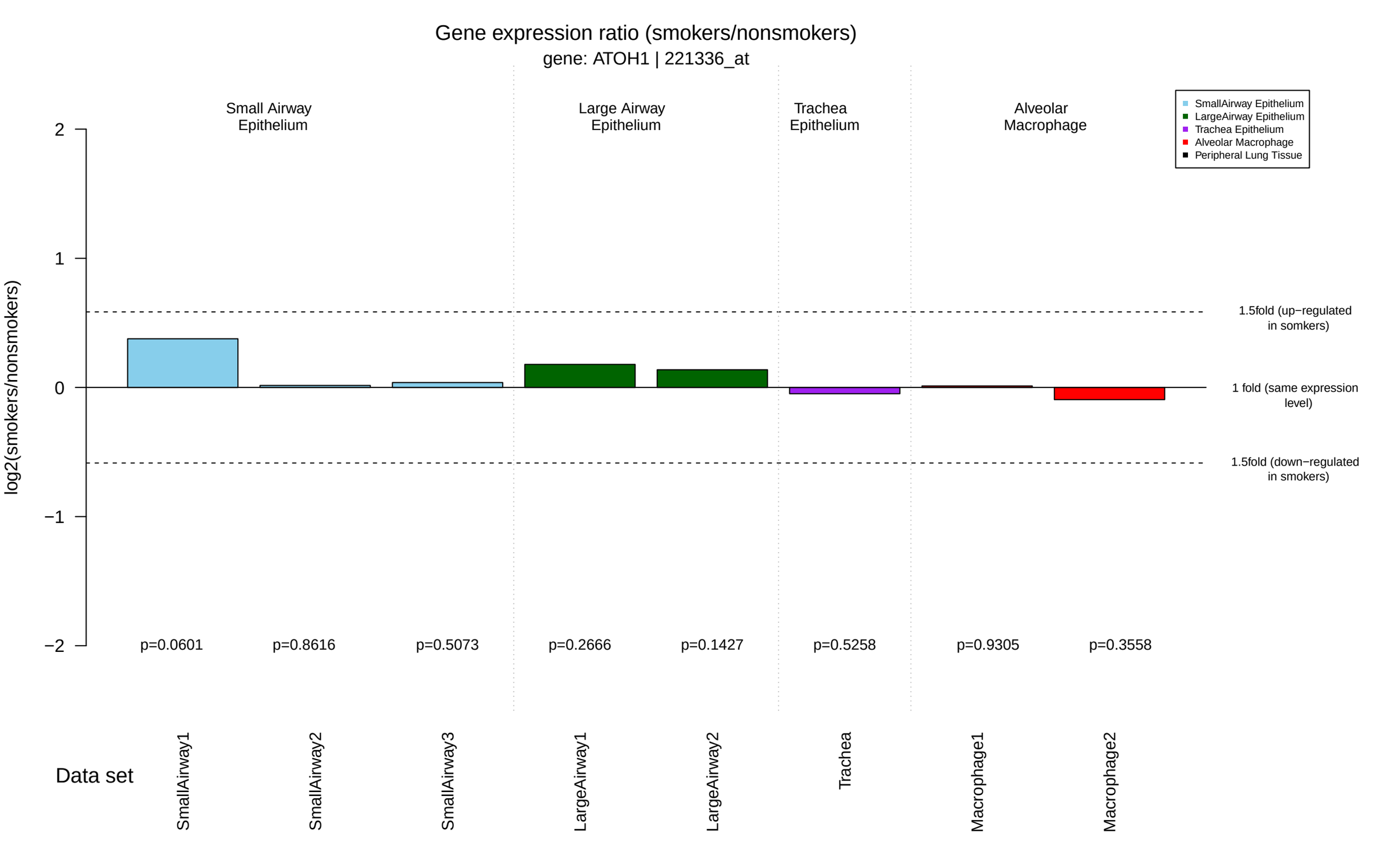
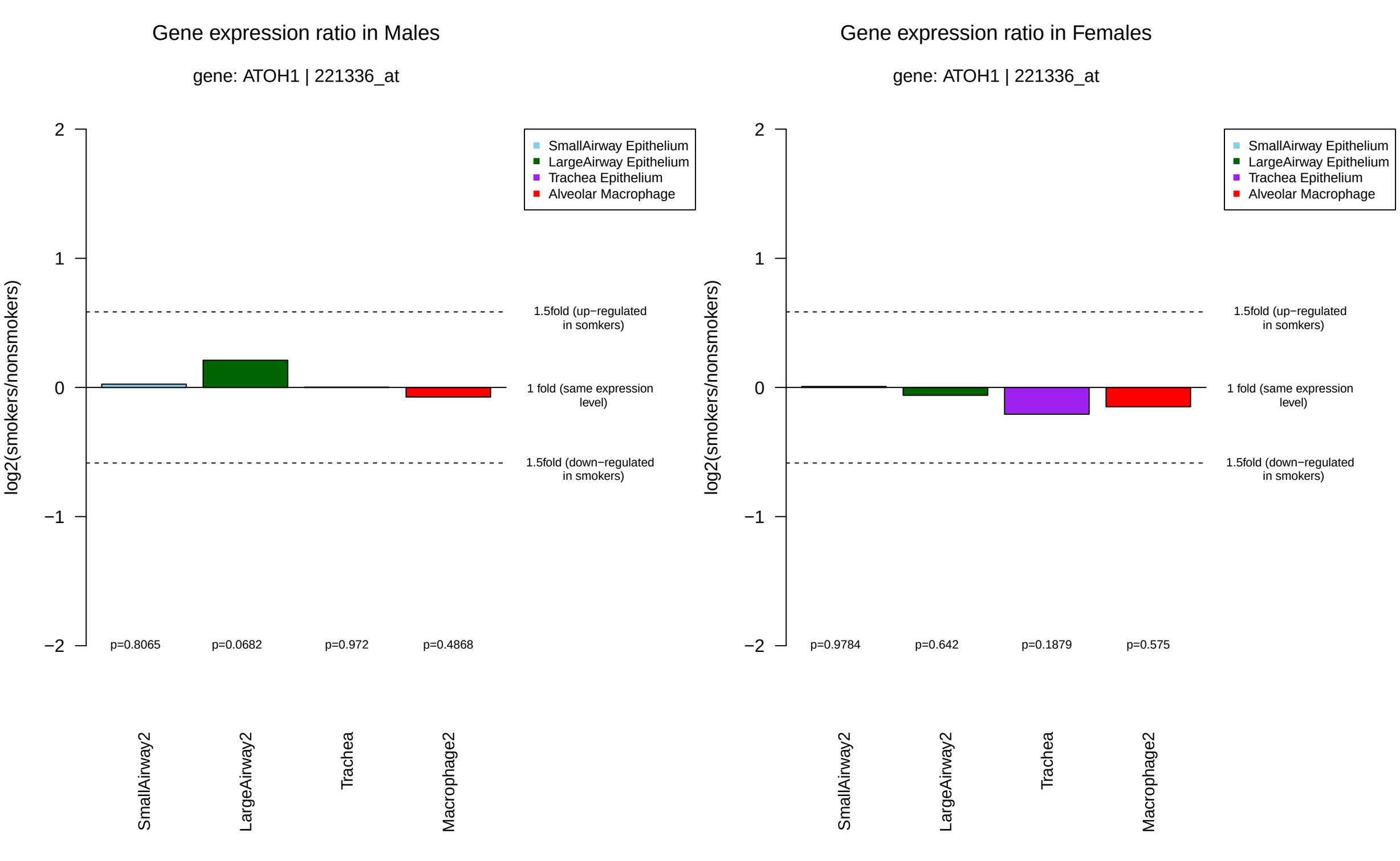
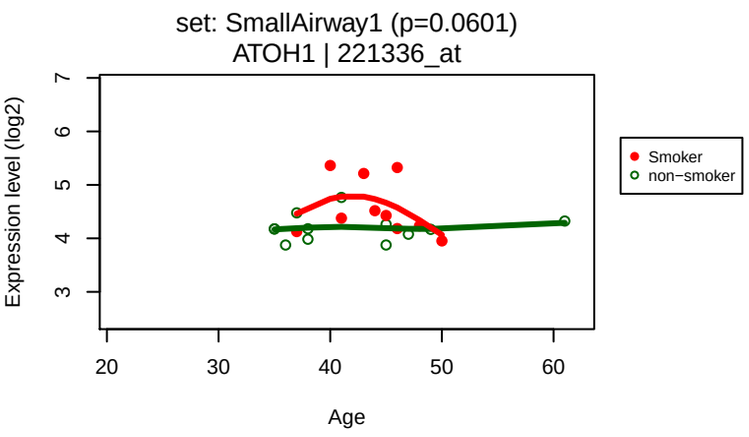
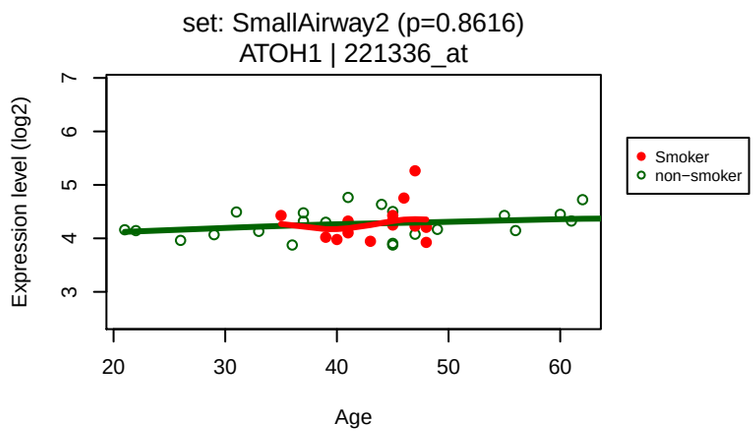
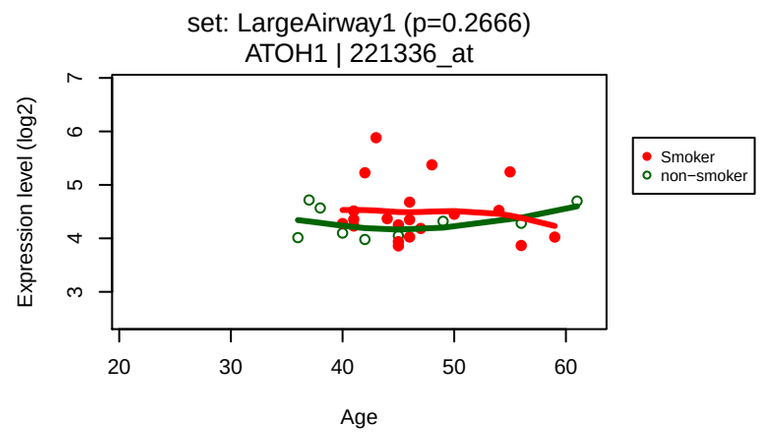

Gene expression of smokers and nonsmokers
gene: ATOH1 | 221336_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0601
p=0.8616
p=0.5073
p=0.2666
p=0.1427
p=0.5258
p=0.9305
p=0.3558
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

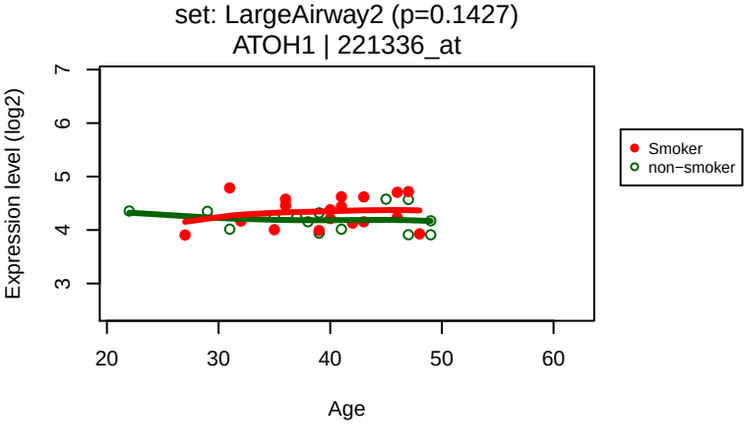

Gene expression ratio (smokers/nonsmokers)
gene: ATOH1 | 221336_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0601
p=0.8616
p=0.5073
p=0.2666
p=0.1427
p=0.5258
p=0.9305
p=0.3558
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

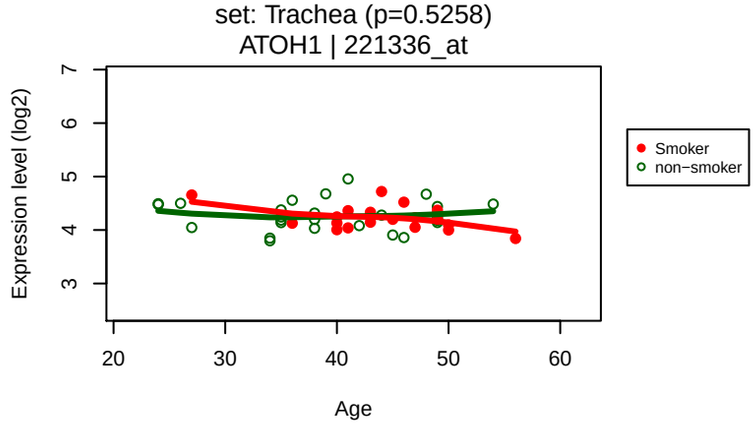

Gene expression ratio in Males
gene: ATOH1 | 221336_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8065
p=0.0682
p=0.972
p=0.4868
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ATOH1 | 221336_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9784
p=0.642
p=0.1879
p=0.575
SmallAirway2
LargeAirway2
Trachea
Macrophage2

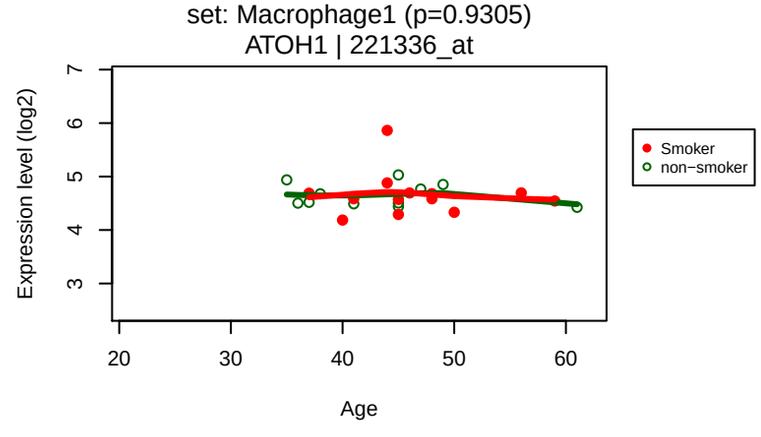

set: SmallAirway1 (p=0.0601)
ATOH1 | 221336_at
Expression level (log2)
Age
Smoker
non−smoker

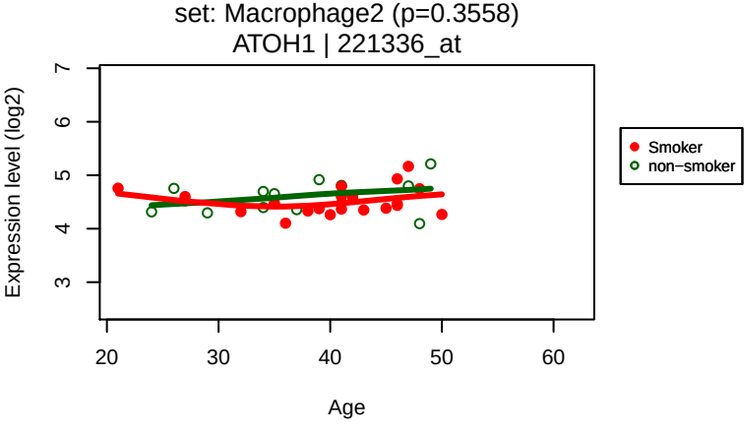

set: SmallAirway2 (p=0.8616)
ATOH1 | 221336_at
Expression level (log2)
Age
Smoker
non−smoker

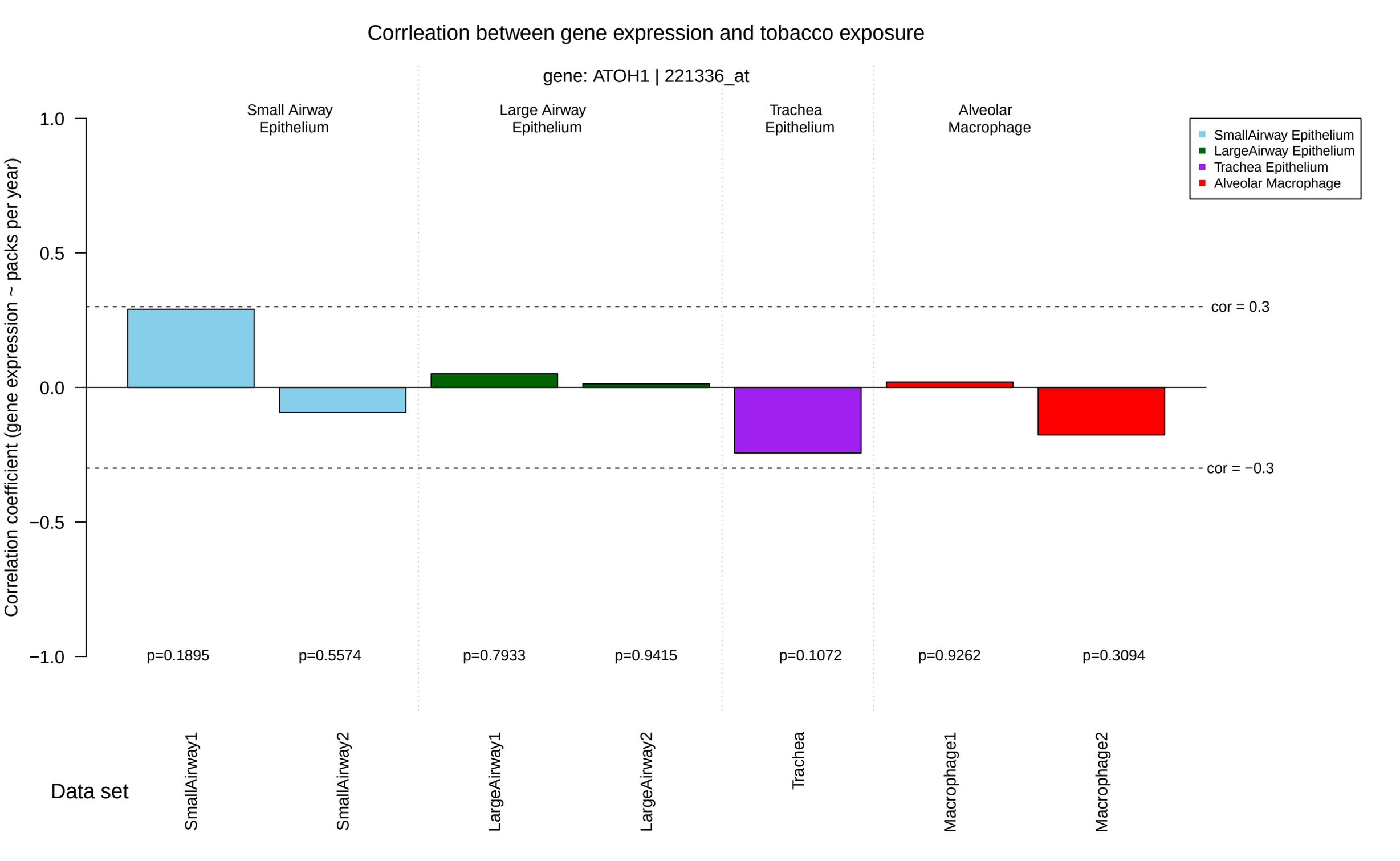

set: LargeAirway1 (p=0.2666)
ATOH1 | 221336_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1427)
ATOH1 | 221336_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.5258)
ATOH1 | 221336_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9305)
ATOH1 | 221336_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3558)
ATOH1 | 221336_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ATOH1 | 221336_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1895
p=0.5574
p=0.7933
p=0.9415
p=0.1072
p=0.9262
p=0.3094
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2